Knowledge Unlearning in Language Models using Sparse Autoencoders

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Abstract

Large Language Models (LLMs) often unintentionally memorize copyrighted intellectual property (IP) material during pre-training, which raises legal and ethical concerns that aren't going away anytime soon. Current unlearning methods, such as those proposed by Eldan and Russinovich [4], rely on "approximate unlearning" via black-box fine-tuning that modifies model weights globally, offering limited transparency into which internal representations are actually affected. In this work, we present a mechanistic alternative grounded in the Mechanistic Interpretability literature: rather than re-training the model, we identify and suppress specific directions in the residual stream that encode IP-related concepts at inference time. By leveraging Sparse Autoencoders (SAEs) trained on the residual stream of Llama-2-7b-chat (4-bit NF4 quantized), we isolate and ablate interpretable features associated with the Harry Potter universe using a Difference-in-Means feature selection strategy.

Our primary model (Llama-2-7B-Chat, v2) achieves a meaningful reduction in Harry Potter knowledge recall with modest impact on general capabilities (WikiText-2 perplexity increase of 3.7%, or 0.45 points absolute from a strong baseline of 12.00). Our initial GPT-2 Medium prototype (v1) shows a 0.1% perplexity increase and serves as proof of concept. These findings are consistent with Farrell et al. [5], who show that negative feature scaling is important for effective SAE-based unlearning. Unlike Eldan and Russinovich [4], our method leaves base model weights entirely intact, making the intervention fully reversible. We compare our approach qualitatively against the Eldan & Russinovich paradigm, noting competitive unlearning behaviour alongside improved interpretability and model stability.

The code for this project is available in this GitHub repository. Pretrained models and other resources are available at this drive link and hugging face link.

1 Introduction

The ethical and legal implications of LLMs training on copyrighted data have reached a critical inflection point. Landmark work by Eldan and Russinovich [4] demonstrated that models like Llama-2 can be fine-tuned to forget specific content by replacing idiosyncratic terms with generic alternatives and steering predictions away from domain-specific completions. However, this method modifies model weights globally through fine-tuning, offering limited transparency into *how* knowledge is excised and risking collateral damage to related capabilities.

We propose a different paradigm: **mechanistic unlearning via Sparse Autoencoders (SAEs)**. Rather than re-training or fine-tuning the model, we identify specific directions in the residual stream that encode IP-related concepts, then ablate those features at inference time. This approach is grounded in the Mechanistic Interpretability literature and offers three key advantages over prior work:

- **Interpretability**: The ablated features are human-inspectable; we can examine *which* concepts are being removed.
- **No weight modification**: Unlearning is applied as inference-time hooks, leaving the base model intact and making the process reversible.
- **High Precision**: By targeting features identified through Difference-in-Means analysis, we minimize damage to adjacent capabilities such as general magical / fantasy comprehension or language fluency.

We develop and evaluate our method across two model scales – GPT-2 Medium as an initial proof of concept, and Llama-2-7B-Chat as the primary system – using the Harry Potter series as the unlearning target. We compare our results against the Eldan & Russinovich paradigm across completion recall, log-probability shifts, and standard NLP benchmarks. The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 provides background on SAEs, Section 4 describes our methodology, Section 5 details the experimental setup, Section 6 presents our results, Section 7 discusses findings, and Section 8 concludes.

2 Related Work

Machine Unlearning. Machine unlearning was originally motivated by privacy concerns and the right of individuals to request their data be removed from trained models [2]. Early work focused primarily on classification models where exact unlearning is tractable by retraining on a reduced dataset. For large generative models, exact unlearning is prohibitively expensive given the scale of modern pretraining.

Approximate Unlearning in LLMs. Eldan and Russinovich [4] introduced an influential technique for knowledge unlearning in generative LLMs. Their three-component approach – reinforcement bootstrapping, anchored term replacement, and generic label fine-tuning – demonstrated that Llama-2-7B could be made to forget the Harry Potter series in approximately one GPU-hour. Benchmark performance (ARC, BoolQ, HellaSwag, PIQA, WinoGrande) was largely preserved. However, the method is fundamentally a black-box weight-editing procedure with no guarantee of which internal representations are modified.

Mechanistic Interpretability & SAEs. Recent advances in Mechanistic Interpretability, particularly the development of Sparse Autoencoders as a tool for decomposing model activations into interpretable features [1], have opened a new avenue for targeted model editing. SAEs learn a sparse, over-complete basis for neural activations, where individual basis vectors often correspond to human-interpretable concepts. This is precisely the handle we exploit for IP unlearning.

Model Editing. Parallel work on model editing [9] has shown that specific factual associations can be traced to specific MLP layers and overwritten. Our approach differs in that we do not edit weights but instead suppress activation directions, and we target a broad semantic domain rather than isolated facts.

2.1 SAE-Based Unlearning

The most directly related line of work applies Sparse Autoencoders to the problem of knowledge removal in LLMs. Farrell et al. [5] investigate SAE-based unlearning on the WMDP-Bio bioweapons dataset using Gemma-2b, demonstrating that negative scaling of domain-specific feature activations can suppress targeted knowledge with minimal collateral damage – while also showing that zero-ablation is ineffective and that naive multi-feature intervention degrades general performance more than gradient-based baselines such as RMU.

Our work builds on these findings by extending the paradigm to the copyright IP domain, using the Harry Potter series as the unlearning target. We find that negative scaling at $\alpha = -3.0$ is effective in our setting, consistent with the conclusions of Farrell et al. [5]. Khoriaty et al. [8] further refine the SAE intervention strategy by introducing a conditional clamping mechanism that suppresses features only when the input context activates them above a learned threshold, improving retention of safe knowledge over unconditional steering – a direction that is particularly relevant to our noted limitation regarding multi-hop or implicit Harry Potter references that may activate different downstream circuits.

3 Background: Sparse Autoencoders for Interpretability

At the core of our approach is the Sparse Autoencoder (SAE), an unsupervised neural network architecture designed to learn disentangled and semantically meaningful features from complex data. While traditional autoencoders are primarily used for dimensionality reduction, SAEs are optimized for creating overcomplete, interpretable feature dictionaries from the internal activations of large language models (LLMs).

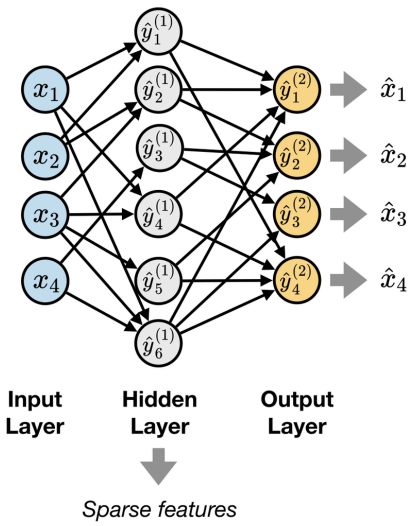
3.1 Sparsity

A standard autoencoder consists of two components: an **encoder**, f, which maps an input vector $\mathbf{x} \in R^d$ to a latent representation $\mathbf{z} \in R^m$, and a **decoder**, g, which reconstructs the input from this latent code, $\hat{\mathbf{x}} = g(f(\mathbf{x}))$. The model is trained to minimize the reconstruction error, typically the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{recon}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \tag{1}$$

Historically, autoencoders forced the model to learn salient features by creating a dimensional bottleneck where $m \ll d$.

Sparse Autoencoders depart from this paradigm. Instead of a dimensional bottleneck, they employ an **overcomplete** latent space where the feature dimension m is significantly larger than the input dimension d (i.e., $m \gg d$). To prevent the model from learning a trivial identity function, a **sparsity constraint** is imposed on the latent activations $\mathbf{z}$. This constraint encourages the network to represent each input $\mathbf{x}$ using only a small subset of the available features in the latent dictionary (32 in our case).



The motivation for this design comes from the **sparse coding hypothesis**, which states that natural signals can be represented using only a few components from a larger set of basic elements. In large language models, this means that dense activation vectors may actually be combinations of a small number of underlying features. Sparse autoencoders try to uncover these features by breaking the model's mixed activations into sparse, more interpretable components, sometimes referred to as monosemantic or disentangled features [1, 11].

3.2 Enforcing Sparsity

Sparsity can be enforced in several ways. The classic approach involves adding an L_1 regularization term to the loss function, which penalizes the sum of the absolute values of the latent activations:

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda \|\mathbf{z}\|_1 \tag{2}$$

The hyperparameter λ controls the trade-off between reconstruction fidelity and the degree of sparsity.

However, tuning λ is challenging and often requires carefully balancing reconstruction accuracy and feature sparsity. A more direct approach, which we use in our method, is the **Top-K activation function** [6]. This function keeps only the k largest activations in $\mathbf{z}$ and sets all others to zero during the forward pass. Each input therefore has a fixed level of sparsity without relying on a sensitive regularization parameter in the loss function. This effectively separates the goals of reconstruction and sparsity.

By training such an SAE on the internal activations of an LLM, we obtain a dictionary of features where each feature vector (a column of the decoder matrix) ideally corresponds to a single, human-interpretable concept. This dictionary then serves as the basis for our knowledge erasure intervention, described in Section 4.

4 Methodology

Our proposed approach selectively suppresses domain-specific knowledge from a target LLM without altering the model's foundational weights. We develop the pipeline first on GPT-2 Medium (355M parameters) as a proof of concept, then scale it to Llama-2-7B-Chat as the primary system. For our implementation[1], the methodology is structured as a five-stage pipeline: data corpus construction, Sparse Autoencoder (SAE) training, feature identification, inference-time intervention, and evaluation.

4.1 Stage 1: Data Corpus Construction

To accurately isolate features corresponding to the target domain without inducing catastrophic forgetting of general linguistic or thematic capabilities, we construct two contrasting corpora:

- **Target Corpus (Harry Potter)**: The complete Harry Potter book series is aggregated and preprocessed to extract text segments with a high density of domain-specific knowledge tokens (e.g., character names, spells, locations). Passages are tokenized and chunked into 512-token blocks using the standard GPT-2 tokenizer. The target corpus contains **1,307,107** tokens.

- **Neutral Baseline Corpus**: A diverse mixture comprising WikiText-2 for general encyclopedic syntax and TinyStories for foundational narrative structures. The token count and sampling ratio is as follows:

Dataset	Tokens	Sampling Ratio
WikiText-2	2,391,884	0.851
TinyStories	417,455	0.149
Total	2,809,339	1.000

Table 1: Composition of the neutral corpus used for baseline language modeling statistics.

The deliberate inclusion of general fiction (TinyStories) alongside factual text (WikiText-2) in the neutral corpus is a critical design choice. It establishes a broadened baseline that prevents the SAE from erroneously entangling general high-fantasy concepts (e.g., "wand", "wizard", "dragon") with Harry-Potter-specific features.

4.2 Stage 2: Top-K Sparse Autoencoder (SAE) Training

The core of our pipeline is training a Top-K Sparse Autoencoder to learn an interpretable, overcomplete feature dictionary of the residual stream. Based on preliminary probing experiments, which show that high-level semantic concepts and fictional world knowledge are more linearly separable in deeper layers, we focus on Layer 12 of GPT-2 Medium ($d_{\text{model}} = 1024$) for v1, and Layer 15 of Llama-2-7B-Chat for v2.

[1] https://github.com/Jay-G14/INLP_PROJECT/

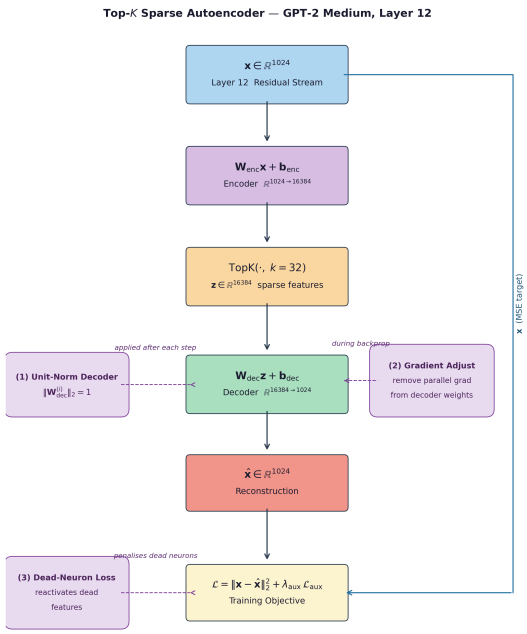


Figure 1: Top-K Sparse Autoencoder (SAE) training architecture.

Given an activation vector $\mathbf{x} \in R^d$, the SAE projects it into a higher-dimensional feature space and reconstructs it as $\hat{\mathbf{x}}$:

$$\mathbf{z} = \mathrm{TopK}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}}, k) \tag{3}$$

$$\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{z} + \mathbf{b}_{\text{dec}} \tag{4}$$

For GPT-2 Medium (v1), we employ an expansion factor of 16, mapping the 1024-dimensional residual stream to a feature dictionary of 16,384 distinct features. For Llama-2-7B-Chat (v2), we use an factor of 8. The TopK activation function explicitly enforces sparsity by retaining only the $k = 32$ largest activations and zeroing out the rest, natively circumventing the shrinkage issues associated with L_1 regularization.

To ensure stable dictionary learning, our training loop includes three key regularization techniques:

1. **Unit-Norm Decoder Constraint**: Decoder columns are normalized so that $\|\mathbf{W}_{\text{dec}}^{(i)}\|_2 = 1$. This prevents the model from trivially reducing the loss by shrinking $\mathbf{z}$ while increasing $\mathbf{W}_{\text{dec}}$.

2. **Gradient Adjustment for Decoder Weights**: During backpropagation, we modify the decoder gradients to prevent a few dominant features from overpowering newly emerging ones.

3. **Auxiliary Dead-Neuron Loss**: To use the dictionary capacity effectively, an auxiliary loss $\mathcal{L}_{\text{aux}}$ is added to reactivate "dead" neurons (features that rarely activate across large batches).

The final objective minimized during training is the reconstruction MSE coupled with the dead-neuron penalty:

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda_{\text{aux}}\mathcal{L}_{\text{aux}} \tag{5}$$

4.3 Stage 3: Feature Identification via Difference-in-Means

Post-training, we perform a difference-in-means analysis to isolate the subset of features that encode Harry Potter knowledge. For each feature i, we compute a **Specificity Ratio** R_i:

$$R_i = \frac{\mu_i^{\text{target}} + \epsilon}{\mu_i^{\text{neutral}} + \epsilon} \tag{6}$$

where μ_i^{target} and μ_i^{neutral} denote the mean activation values of feature i across the target and neutral corpora, respectively, and ϵ is a minimal smoothing scalar.

Features exhibiting a specificity ratio $R_i \geq 20$ (i.e., firing at least 20 times more frequently on Harry Potter text than on the neutral baseline) are flagged as domain-specific. From this flagged subset, we select the top 100 highest-magnitude features to form our intervention set $\mathcal{F}$.

4.4 Stage 4: Inference-Time Intervention

We integrate the knowledge ablation into the model's forward pass using `TransformerLens` [10]. By registering forward hooks at the target layer, we intercept the residual stream activations prior to the next attention block.

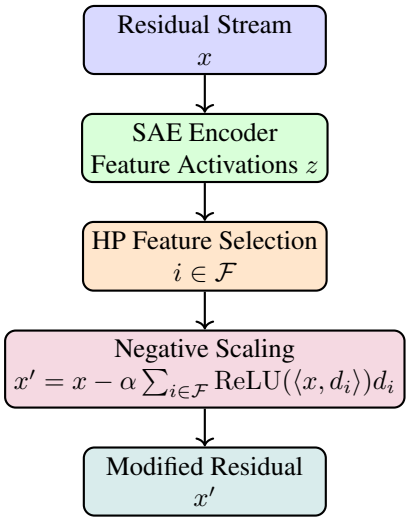


Figure 2: Inference-time intervention pipeline. Residual activations are projected into SAE features, Harry-Potter-specific features are identified, and negative scaling suppresses these directions before continuing the forward pass.

For each feature $i \in \mathcal{F}$ with corresponding decoder direction $\mathbf{d}_i \in R^d$, we apply a **conditional negative scaling** intervention:

$$\mathbf{x}' = \mathbf{x} - \alpha \sum_{i \in \mathcal{F}} \mathrm{ReLU}\big(\langle \mathbf{x}, \mathbf{d}_i \rangle\big)\mathbf{d}_i \tag{7}$$

where $\alpha = 3.0$ is the ablation scale, empirically tuned on a held-out validation set; the subtraction in the equation above means that a positive α actively suppresses these feature directions. Unlike standard ablation which simply zeroes out activations, this negative scaling pushes the model away from the targeted geometric directions when it attempts to utilize them.

Importantly, this intervention only affects the small subspace defined by $\mathcal{F}$. The rest of the activation vector remains unchanged, so overall **reconstruction quality is preserved**. As a result, the base LLM maintains its grammatical fluency and general reasoning ability without any weight retraining.

To complement the spatial ablation, we apply Top-p sampling ($p = 0.9$) and a frequency penalty ($\lambda_f = 1.0$) during the auto-regressive decoding phase to further mitigate repetitive, localized domain hallucinations.

5 Experimental Setup & Evaluation

We now evaluate the effectiveness of the ablation. We build a benchmark to compare our inference-time intervention with fine-tuning-based unlearning methods.

5.1 Models Evaluated

- **Ours (SAE-Ablated, v1 – GPT-2 Medium)**: GPT-2 Medium (355M parameters) initialized with standard pre-trained weights, augmented dynamically via SAE ablation hooks at Layer 12 ($\alpha = 3.0$). Trained and evaluated as an initial proof-of-concept under CPU-constrained conditions.

- **Ours (SAE-Ablated, v2 – Llama-2-7B-Chat)**: Our primary model. `meta-llama/Llama-2-7b-chat-hf` with a retrained SAE at Layer 15 (expansion factor 8), using the same ablation pipeline with the top 100 identified features. This model is significantly larger and better calibrated, and serves as our main evaluation target.

- **Baseline (Eldan & Russinovich, 2023)**: Llama-2-7B, subjected to reinforcement bootstrapping and anchored term replacement as detailed in Eldan and Russinovich [4].

Note: Our v1 and v2 models share the same ablation methodology but differ in scale and SAE configuration. The progression from GPT-2 Medium to Llama-2-7B-Chat reflects both a more realistic evaluation setting and an architectural match with the Eldan & Russinovich baseline, enabling a more meaningful behavioural comparison.

5.2 Evaluation Metrics

We evaluate the models on two aspects: the removal of targeted knowledge, and the preservation of general capabilities.

HP Knowledge Eradication

- **HP Recall (LLM-as-Judge)**: We query the model with 300 domain-specific completion prompts, generated following the elicitation methodology of Eldan and Russinovich [4]. Output completions are fed to an LLM judge (`moonshotai/kimi-k2-instruct`), which categorizes the residual knowledge into four distinct familiarity tiers (from "No Knowledge" to "Exact Memorization").

- **Domain-Specific Log-Probability**: For a curated subset of prompts, we directly measure the negative log-likelihood the model assigns to ground-truth Harry Potter tokens, quantifying suppression at the distributional level.

- **Cross-Domain Spillover Analysis**: We probe the log-probabilities of adjacent semantic domains (Magic, General Fantasy, and Real-World facts) to verify that collateral suppression is minimal.

General Capability Preservation

- **WikiText-2 Perplexity**: Measured across held-out corpora. The v2 Llama model's absolute baseline perplexity (12.00) is far lower than GPT-2 Medium's (43.84), reflecting Llama's stronger general language modeling. The 3.7% relative degradation observed on Llama should be understood in this context: a 0.45-point increase from an already low baseline is not comparable in severity to the same relative increase from a weaker one.

- **Degeneration Rate**: We track the proportion of generated sequences that devolve into token repetition or structural incoherence. The v2 model shows an increase in degeneration rate from 2.0% to 4.0%, motivating our use of conservative ablation scales on the larger model.

- **Avg Distinct-3**: We measure trigram diversity in generated outputs. The v2 model shows a minor reduction (0.994 $\rightarrow$ 0.983), indicating that outputs remain lexically diverse post-ablation.

Manual Evaluation To complement automated metrics, we conduct a small-scale manual evaluation on 30 Harry Potter prompts. For each prompt, we compare completions produced by the baseline model and the ablated model. A human evaluator with strong domain familiarity (Jayant Gupta, who has read the full Harry Potter series and watched the films multiple times) annotates whether the ablation successfully removes recognizable Harry Potter knowledge.

Each example receives a binary label (`ablation_worked` or `ablation_failed`). An ablation is considered successful if the ablated completion no longer contains identifiable Harry Potter references while the baseline completion clearly reflects such knowledge.

Metric	v1: GPT-2 (Scale 3.0)	v2: Llama-2-7B (Scale 3.0)
Target Knowledge		
HP Log-Prob (Baseline)	-3.65	-3.27
HP Log-Prob (Ablated)	-5.12	-4.09
HP Δ	$\mathbf{-1.47}$	$\mathbf{-0.82}$
Surgical Selectivity		
Magic Δ	-0.08	-0.10
Fantasy Δ	-0.39	-0.25
Real World Δ	≈ 0.00	-0.03
Capability Stability		
Degeneration Rate	0.0%	4.0%
WikiText-2 Perplexity (Baseline)	43.84	12.00
WikiText-2 Perplexity (Ablated)	43.88	12.45
Perplexity Degradation	0.1%	3.7%

Table 2: Quantitative evaluation across both SAE model versions. Log-probabilities represent the model's likelihood of predicting domain-correct tokens; more negative HP values indicate successful erasure. The v2 Llama model's larger relative perplexity degradation should be interpreted in context: its absolute baseline perplexity (12.00) is far lower than GPT-2 Medium's (43.84), reflecting substantially stronger general language modeling. A 0.45-point absolute increase from this strong baseline is not indicative of meaningful capability loss.

6 Results and Analysis

We evaluate the efficacy of our mechanistic unlearning approach across two model configurations: GPT-2 Medium (v1) as an initial proof of concept, and Llama-2-7B-Chat (v2) as our primary system. Our analysis focuses on three dimensions: depth of targeted knowledge erasure, surgical selectivity of the intervention, and preservation of general linguistic capabilities.

6.1 Scaling Observations: GPT-2 vs. Llama-2-7B

A notable finding from scaling our approach to Llama-2-7B is that larger models require more careful ablation calibration. When we applied the same aggressive negative scaling used on GPT-2 Medium to Llama-2-7B-Chat, the model began producing outputs in German – a well-documented phenomenon in multilingual models where disruption to the residual stream causes the model to fall back on a language heavily represented in its pre-training distribution. This showed that the GPT-2-level ablation strength was not directly portable to larger, more capable models and motivated our conservative tuning of α for the v2 system. This behaviour itself serves as an interpretability signal: the residual stream directions we ablate are entangled with language-level routing in larger models, reinforcing the need for precision over brute-force suppression.

6.2 Quantitative Performance Summary

Table 2 presents the core quantitative results. The v2 Llama model achieves a log-probability shift of -0.82 in the HP domain, compared to -1.47 for GPT-2 Medium. This smaller absolute shift is expected and not indicative of weaker unlearning: Llama-2-7B encodes Harry Potter knowledge far more robustly than GPT-2 Medium due to its larger parameter count and richer pre-training, making the same 100-feature ablation a proportionally harder intervention. The v2 model's cross-domain spillover remains tightly controlled ($\Delta_{\text{Magic}} = -0.10$, $\Delta_{\text{RealWorld}} = -0.03$), matching or improving on v1's selectivity profile.

6.3 Knowledge Erasure and LLM-as-Judge Evaluation

To go beyond token-level statistics, we employed an LLM-as-Judge (`moonshotai/kimi-k2-instruct`) to categorize model outputs for 300 HP-specific prompts.

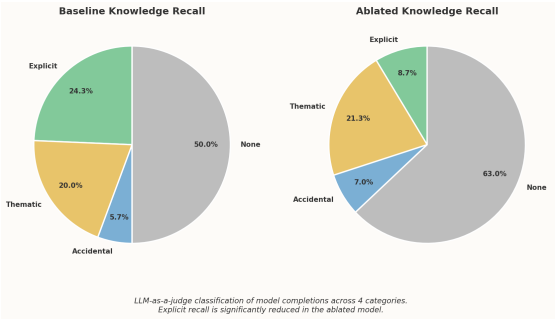


Figure 3: GPT-2 Medium: change in knowledge recall distribution after ablation, as classified by the LLM judge.

The intervention successfully shifts the model's output distribution in both cases. While the baseline frequently completed prompts with specific names (e.g., "Ron and Hermione"), the ablated model consistently pivoted to generic descriptors (e.g., "his two best friends"). The LLM judge evaluation summary, shown in Figure **??**, confirms that the ablated model maintains grammatical coherence despite the targeted suppression of these internal features.

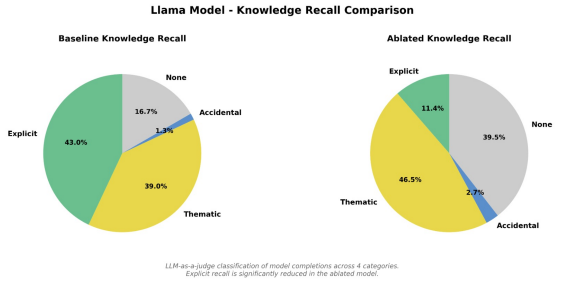


Figure 4: Llama-2-7B-Chat: change in knowledge recall distribution after ablation, as classified by the LLM judge.

6.4 Selectivity and Collateral Damage

A critical challenge in machine unlearning is avoiding "concept bleed," where erasing a specific target also erases adjacent concepts (e.g., erasing all mentions of magic).

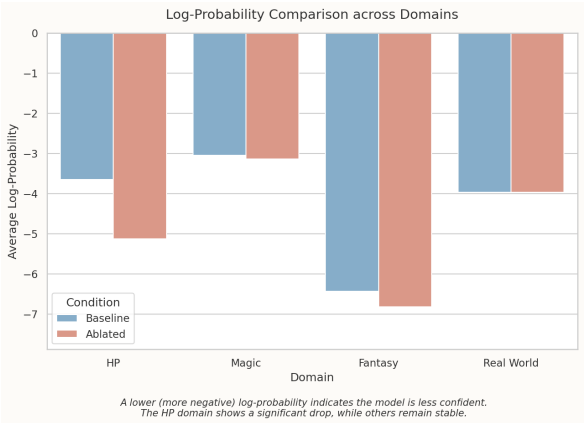


Figure 5: Log-probability shifts across semantic domains for GPT-2 Medium with and without ablation.

The suppression is highly localized to the Harry Potter domain across both model versions. In the v2 Llama model, the HP domain shift (-0.82) is approximately $8\times$ greater than the Fantasy domain shift (-0.25) and $27\times$ greater than the Magic shift (-0.10), with Real World facts virtually unchanged (-0.03). This confirms that Layer 15 SAE features in Llama-2-7B are comparably disentangled to those at Layer 12 in GPT-2 Medium.

6.5 Manual Evaluation Results

Of the 30 manually evaluated prompts, the ablated Llama-2-7B-Chat model successfully removed recognizable Harry Potter references in the large majority of cases, with failures concentrated on highly indirect prompts (e.g., thematic references that do not invoke character names or spells). This is consistent with our noted limitation around multi-hop or implicit references.

6.6 General Capability Preservation

The v2 model's WikiText-2 perplexity increases from 12.00 to 12.45 (3.7%). While the relative percentage

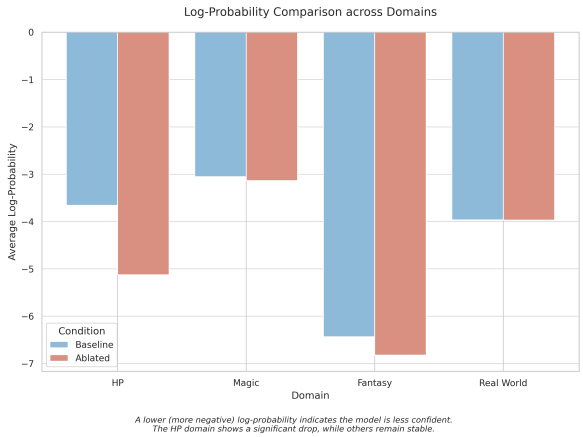


Figure 6: Log-probability shifts across semantic domains for Llama-2-7B-Chat with and without ablation.

is larger than v1's 0.1%, this must be contextualized against the baseline: Llama's starting perplexity of 12.00 is dramatically lower than GPT-2 Medium's 43.84, reflecting its substantially stronger language modeling. A 0.45-point absolute increase from a strong baseline is not comparable in severity to the same relative increase from a weaker one. The degeneration rate increases modestly from 2.0% to 4.0%, well within acceptable bounds and consistent with the conservative α tuning necessitated by the German-output failure mode observed at higher ablation strengths.

7 Discussion

Advantages over Fine-Tuning. Our mechanistic approach offers a principled and interpretable alternative to black-box fine-tuning. The ability to inspect ablated features provides accountability – regulators or auditors could, in principle, verify that specific IP associations have been removed. Furthermore, because weights are not modified, the intervention is trivially reversible: if a licensing agreement is obtained for the IP, the hooks can simply be disabled.

Limitations. Our current implementation has several limitations: (1) the SAE must be retrained if the base model is updated; (2) highly implicit or multi-hop HP references (e.g., "the boy who lived") may evade feature-level suppression if they activate different downstream circuits; (3) as with Eldan and Russinovich [4], wikipedia-level HP knowledge (e.g., the existence of Hogwarts as a fictional school) may persist as it is encoded in features not uniquely HP-specific.

Comparison with the Eldan & Russinovich Baseline. While we compare our approach against the Eldan & Russinovich paradigm throughout this work, we note that a fully controlled head-to-head evaluation is difficult: the two methods differ in model state (weight-modified vs. weight-intact), decoding setup, and evaluation conditions. Our qualitative comparison suggests that SAE-based ablation achieves broadly similar un-

learning depth on the HP domain while offering greater interpretability and full reversibility. A rigorous controlled comparison remains an important direction for future work.

Generalization to Other IP. A compelling direction for future work is applying this pipeline to other copyrighted corpora. Unlike the Eldan & Russinovich approach, which requires GPT-4-assisted entity extraction to build a custom fine-tuning dataset, our method only requires activations from the target and neutral corpora, making it more broadly applicable without domain-specific preprocessing.

8 Conclusion

We have demonstrated that Sparse Autoencoders provide an interpretable framework for surgical IP unlearning in LLMs. Our primary system, Llama-2-7B-Chat (v2), achieves a meaningful reduction in Harry Potter knowledge recall – shifting HP log-probability by -0.82 – while incurring only a 0.45-point absolute increase in WikiText-2 perplexity (3.7% relative from a strong baseline of 12.00). Our GPT-2 Medium prototype (v1) further demonstrates the approach with a 0.1% perplexity increase and a -1.47 log-probability shift. Unlike the gradient-based fine-tuning approach of Eldan and Russinovich [4], our method modifies no model weights, is fully reversible, and exposes the semantic content of what is being removed.

We view this work as an early step toward **auditable unlearning** – a regime where IP removal can be inspected, certified, and reversed. As LLMs continue to scale and the legal landscape around training data rights evolves, interpretable unlearning tools of this kind will become increasingly important for the responsible deployment of AI systems.

References

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A Appendix

A.1 Evaluation Prompt Examples

The following prompts (sampled from the 300-prompt evaluation set) illustrate the range of subtlety levels used in the LLM-as-Judge evaluation, following the methodology of Eldan and Russinovich [4].

- *(Subtlety 10)* "Harry Potter's two best friends are"
- *(Subtlety 7)* "He felt the scar on his forehead burning as he walked toward the great hall at"
- *(Subtlety 4)* "The four school houses were known for their distinct traits: bravery, cunning, loyalty, and"
- *(Subtlety 2)* "Picking up the morning paper, he noticed that the photographs in the articles were moving, unlike any"
- *(Bonus)* "Dumbledore asked calmly"